

WAVELENGTH, ABSORPTION, DIFFRACTION, REFRACTION

- WE HAVE TOUCHED THE IDEA OF WAVELENGTH,
SO NOW LET'S FORMALIZE IT AND DISCUSS ITS CONSEQUENCES
 - PERIOD, WAVELENGTH, FREQUENCY, SPATIAL FREQUENCY, & WAVE SPEED
 - DIFFRACTION
 - LOSS
 - REFLECTION & TRANSMISSION
 - REFRACTION

WAVELENGTH

DEMO PHET WAVE ON A STRING

- NO END
- NO DAMPING
- SLOW MOTION
- OSCILLATE

- WAVES OSCILLATE IN SPACE AND TIME. LET'S REVISIT THIS MORE CAREFULLY....

- WAVES ARE CHARACTERIZED BY:

1. PERIOD (T): TIME BETWEEN CYCLES

- AT FIXED LOCATION, WE'RE "USED" TO SEEING THE GENERAL PURE TONE

$$y(t) = y_0 \cos\left(2\pi \frac{t}{T} + \phi_1\right)$$

↑ CONSTANT PHASE (FIXED VALUE OF x)
↑ AMPLITUDE
↑ WHEN t CHANGES BY T, COSINE REPEATS!
↑ COULD BE y, OP, OP, T, ...

2. WAVELENGTH (λ): SPACE BETWEEN CYCLES

- A SNAPSHOT IN TIME:

$$y(x) = y_0 \cos\left(\phi_2 + 2\pi \frac{x}{\lambda}\right)$$

↑ CONSTANT PHASE (FIXED VALUE OF t)
↑ WHEN x CHANGES BY λ, COSINE REPEATS!

- IF WE ALLOW TIME VARIATION AT EACH POSITION (OR VIS VERSA)

$$\phi_2 \rightarrow 2\pi \frac{t}{T} \quad \left(\phi_1 \rightarrow 2\pi \frac{x}{\lambda}\right)$$

⇒ A TRAVELLING WAVE FOR A PURE TONE:

$$y(x,t) = y_0 \cos\left(2\pi \frac{t}{T} + 2\pi \frac{x}{\lambda} + \phi_0\right)$$

DEMO DESMOS PLOT WITH SLIDER FOR t, T, λ, φ₀.

- PLAY w/ EACH AFTER GUESSING OUTCOME
- ANIMATE + DO THE SAME AFTER GUESSING WHAT HAPPENS TO WAVE SPEED.
- NEGATIVE t OR x

↑ INCLUDED THIS "OFFSET" CONSTANT, DEPENDS ON DEFINITION OF t=0 & x=0 (MAKES THIS FULLY GENERAL, COVERS SIN() AS WELL)

END OF
W6D OCT 2

10pt

Q WHAT IS THE WAVE SPEED FOR THIS?

- WE CAN ALSO TALK ABOUT REPETITION RATES (INSTEAD OF TIME + SPACE)

$\xrightarrow{\text{CYCLES PER SECOND}} f \equiv \frac{1}{T}$ IN TIME (FREQUENCY, UNITS $\frac{1}{s} = \text{Hz}$)
 $\xrightarrow{\text{CYCLES PER METER}} k \equiv \frac{1}{\lambda}$ IN SPACE ("SPATIAL FREQUENCY", UNITS $\frac{1}{m}$)
 $\equiv \text{"k" OR "WAVENUMBER"}$

$$\Rightarrow y(x, t) = y_0 \cos(2\pi f t + 2\pi k x + \phi_0)$$

- AND $c = \frac{\lambda}{T} = \frac{f}{k} = \lambda f = \dots$

^{20pt}
Q WHAT IS ϕ_0 IF
 $y(0, 0) = y_0$
 $y(0, 0) = 0$

- TO SUMMARIZE:

$$c = \frac{\lambda}{T} = \lambda f = \frac{f}{k} = \frac{1}{Tk}$$
 WAVE SPEED (JUST USE UNITS!)
 $\xrightarrow{\text{WAVELENGTH [m]}}$ $\xrightarrow{\text{PERIOD [s]}}$ $\xrightarrow{\text{FREQUENCY [Hz]}}$ $\xrightarrow{\text{SPATIAL FREQUENCY [$\frac{1}{m}$]}}$

DEMO DESMOS WAVE WITH HARMONICS
 - 5.1 WAVELENGTH FIRST PLOT

^{10pt}
Q WHAT IS THE WAVELENGTH?

- SWITCH TO SECOND PLOT (IRRATIONAL FREQUENCIES)

^{10pt}
Q WHAT IS THE WAVELENGTH?

DIFFRACTION

HOW SOUND SPREADS AROUND OBJECTS

- YOU EXPERIENCE DIFFRACTION EVERY DAY, HEARING SOUND AROUND OBJECTS
- IF I BLOCK BY MOUTH FROM VIEW, YOU CAN STILL HEAR ME TALKING, BECAUSE SOUND IS A WAVE AND ALL WAVES DIFFRACT AROUND BARRIERS

(LIGHT IS A WAVE, SO WHY CAN'T YOU SEE AROUND MY HAND?)

- DO YOU HEAR A CHANGE IN TIMBRE WHEN I BLOCK IT? (YOU BETTER!)

PURE TONES

DEMO PHET WAVE INTERFERENCE - SLITS

- SELECT "SOUND" (SPEAKER)

- HERE THE YELLOW WALLS ARE PERFECT SOUND ABSORBERS THAT THEMSELVES WILL NOT VIBRATE. WHAT WILL HAPPEN IF WAVES LAND ON THIS FROM THE LEFT?

- TURN ON SOUND GENERATOR

- WAVES SPREAD AROUND BARRIERS!

- ALL WAVES DO THIS FOR A VERY DEEP REASON (SEE HUYGEN'S PRINCIPLE)

- SOUND

- WATER

- LIGHT ← (DEMO WITH LASER POINTER & NAIL CLIPPER SLIT)

- QUANTUM PARTICLES (INCLUDING YOU!)

- WHY DOES SOUND DO THIS SO WELL COMPARED TO LIGHT, e.g.?

- YOU CAN SEE HOW THE WAVE GOES EVERYWHERE BUT WITH DIFFERENT INTENSITY...

- IN PHET, MOVE SLIT LEFT & USE PRESSURE SENSOR TOOL TO FIND THE DIRECTION OF $\sim \frac{1}{2}$ AMPLITUDE

- WHAT WILL HAPPEN IF λ GETS SHORTER?

- MAX OUT FREQUENCY & SHOW YOU MUST MOVE THE SENSOR CLOSER TO THE OTHER SENSOR TO MAINTAIN $\frac{1}{2}$ AMPLITUDE

10pt



Q GUESS WHAT HAPPENS IF THE SLIT GETS SMALLER?

- SHOW THIS (MIN SLIT)

⇒ WHEN SLIT WIDTH $\ll$ WAVELENGTH, LOOKS LIKE A POINT SOURCE
WHEN SLIT WIDTH $\gg$ WAVELENGTH, LOOKS MORE LIKE THE WALLS CAST A SHADOW

- SHOW THIS (MAX SLIT, MAX FREQUENCY)

- FOR SCALE:

TYPICAL SOUND $f \sim 300\text{Hz} \Rightarrow \lambda \sim 1\text{m}$

TYPICAL LIGHT $f \sim 500\text{THz} \Rightarrow \lambda \sim 600\text{nm}$ (RED)

SOUND

DOOR SIZE.

LIGHT

HAIR DIAMETER $\div 100$

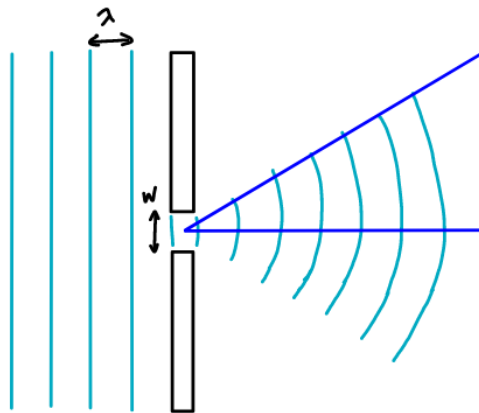
- DOORS CAST SHADOWS IN LIGHT

- DOORS DIFFRACT SOUND

- MY HAND $\ll$ DOOR DOES $\sim$ NOTHING (THOUGH YOU MIGHT HEAR THE HIGHER FREQUENCIES DECREASE INTENSITY)

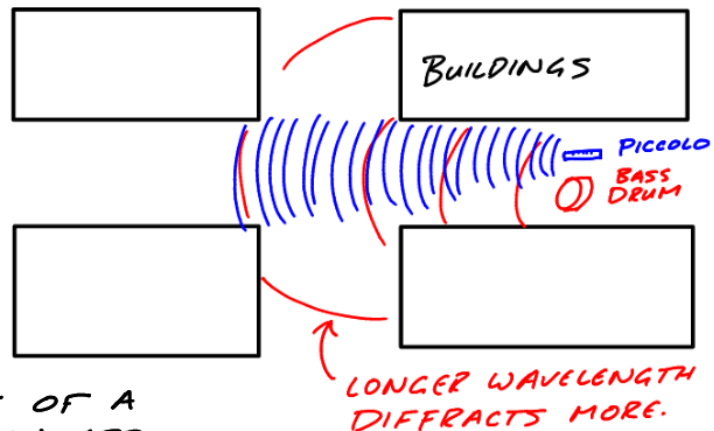
THIS IS PARTLY WHY YOUR EAR + (TRAINED) BRAIN CAN INFER DIRECTIONALITY WITH ONLY ONE EAR
- HELPFUL FOR SPATIAL / 3D AUDIO!

- IT TURNS OUT THAT THE SPREAD ANGLE



WAVELENGTH
 $\theta_{\text{slit}} \approx \frac{\lambda}{w}$
SPREAD ANGLE FOR PLANE WAVES LEAVING A CONFINED SOURCE (DIFFRACTION ANGLE)
WIDTH OF SOURCE / CONFINEMENT
 $\pm \theta_{\text{slit}}$ CONTAINS MOST OF THE POWER

- THIS IS WHY YOU HEAR THE BASS DRUMS OF A MARCHING BAND FIRST



- EVEN THE SHAPE OF A CONCERT HALL CAN AFFECT THE PERCEIVED BALANCE OF INSTRUMENTS! (MORE LATER)

- SO CAN THE SHAPE OF THE PEOPLE & INSTRUMENTS!

Q HOW SHOULD WE ARRANGE AN ORCHESTRA?

COMPLEX TONES SO HOW DO WE PREDICT HOW A NON-SINUSOIDAL WAVE WILL PROPAGATE?

- RECALL ANY WAVE $\Delta P(t)$ CAN BE DECOMPOSED TO (WRITTEN AS A SUM OF) PURE TONES. INCLUDING SPATIAL DEPENDENCE, WE CAN WRITE (1D)

$$\Delta P(x,t) = \Delta P_1 \cos(2\pi f_1 t + 2\pi \kappa_1 x + \phi_1) + \Delta P_2 \cos(2\pi f_2 t + 2\pi \kappa_2 x + \phi_2) + \dots$$

- EACH PURE TONE HAS A WELL-DEFINED WAVELENGTH
- RECALL "LINEARITY" $\Rightarrow$ THESE PURE TONE WAVES BEHAVE INDEPENDENTLY

$\Rightarrow$ PREDICTION METHOD (USEFUL FOR MORE THAN THIS!)

THIS IS SO POWERFUL. IT IS WHY FOURIER SERIES ARE SO INSANELY USEFUL!

- ① DECOMPOSE WAVE INTO PURE TONES
- ② PREDICT HOW EACH EVOLVES (SIMPLER PROBLEM!)
- ③ RECOMBINE TO GET THE FULL WAVE.

- CONSIDER A COMPLEX TONE $\Delta P_{\text{SOURCE}} = A_1 \cos(2\pi f_1 t) + A_2 \cos(2\pi f_2 t)$ $f_2 > f_1$

DEMO DESMOS 5.2 COMPLEX TONE
- JUST FIRST PLOT

- SUPPOSE THIS TONE IS EMITTED AT A SLIT AS BELOW



10pts
Q WILL THE WAVEFORM AT THE EAR HAVE THE SAME SHAPE?

ALSO:

- NO NEW FREQUENCIES ARE GENERATED
- EVERY FREQUENCY DIFFRACTS DIFFERENTLY

$\Rightarrow$ MUST LOOK LIKE SUM OF PURE TONES $f_1 + f_2$

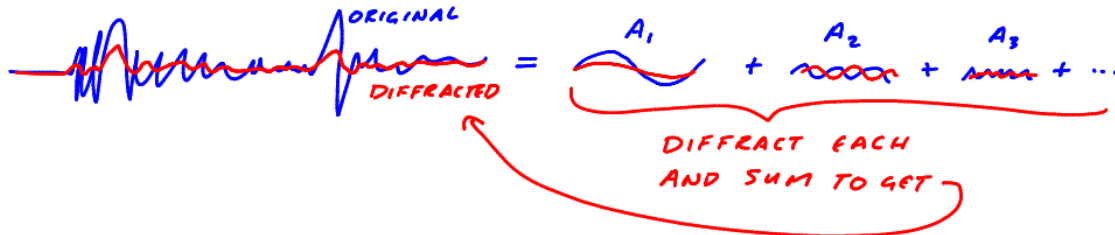
DETERMINED BY DIFFRACTION + PROPAGATION

$$\Rightarrow \Delta P_{\text{EAR}} = B_1 \cos(2\pi f_1 t + \phi_1) + B_2 \cos(2\pi f_2 t + \phi_2)$$

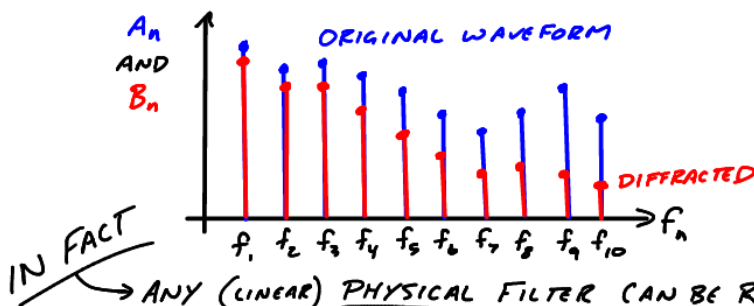
(YOU ARE NOT RESPONSIBLE FOR CALCULATING $\phi_1 + \phi_2$ IN THIS COURSE)

- SHOW 2nd TWO CURVES (COMPONENTS)
- "FILTER" THESE W/ DIFFRACTION, LETTING $A_1 \rightarrow A_1/2$ & $A_2 \rightarrow A_2/4$ (TYPE INTO EXPRESSION, NO SLIDER)
- CHANGE DISTANCE TO SEE PHASES
- PLOT SUM & ORIGINAL ALONE
- "FILTERED" BY DIFFRACTION

- WE CAN IN PRINCIPLE DO THIS FOR ARBITRARY WAVEFORMS



- IN TERMS OF SPECTRA,



- THE FREQUENCY-DEPENDENT RATIO B_n/A_n (TOGETHER WITH ϕ_n) IS CALLED A

"FILTER" OR "TRANSFER" FUNCTION

ANY (LINEAR) PHYSICAL FILTER CAN BE REPRESENTED/SIMULATED WITH AN AMPLITUDE (AND PHASE) AT EACH FREQUENCY:

- TALKING INTO A JAR
- INSTRUMENT RESONANCES
- LIVE SOUND

BONUS: SIMULATE DIFFRACTION FILTER

DEMO

OPEN MARCHING BAND CLIP IN AUDACITY

- SELECT ALL, EFFECT $\rightarrow$ GRAPHICAL EQ



- THIS WILL DO THE ABOVE: THE CURVE IS JUST $\frac{B_n}{A_n}$ (dB)
- ITERATIVELY PREVIEW & INCREASE "KNEE" FREQUENCY.
- AROUND 5 kHz IT SOUNDS LIKE YOU CAN "SEE THE HORNS"

- THIS IS HOW THE SHAPE OF YOUR EAR CAN HELP DISCERN THE DIRECTION OF A SOUND...



BONUS: CONSTRUCT AN EAR-SHAPED MICROPHONE AND MEASURE THIS $\hat{}$ (i.e. DESIGN SPATIAL AUDIO)

ABSORPTION

DURING PROPAGATION

- ALL MATERIALS (INCLUDING AIR) ABSORB SOME SOUND AS IT PASSES
 - USUALLY CONVERTED TO HEAT
 - THIS IS FREQUENCY-DEPENDENT IN GENERAL
 - USUALLY HIGH FREQUENCIES ARE ABSORBED MORE/FASTER
 - THIS IS MULTIPLIED BY ANY "GEOMETRICAL" INTENSITY CHANGE (E.G. $1/r^2$)
- ABSORPTION DURING PROPAGATION IS SPECIFIED BY INTENSITY LOSS PER M (USUALLY IN dB/M) $\leftarrow$ "LOSS RATE" IN dB/M

$$I(x) \propto 10^{-\alpha x/10}$$

LOSS DECAY FACTOR

$\leftarrow$ DISTANCE TRAVELED

- ENTHUSIASTS {
- ARISES FROM THE FACT THAT THINGS LIKE THE AMOUNT OF HEAT GENERATED BY A WAVE ARE PROPORTIONAL TO THE AMPLITUDE OF A WAVE.
 - AS USUAL IT IS EASIEST TO CONSIDER ONE PURE TONE AT A TIME, THEN CONSTRUCT THE FULL WAVEFORM FROM PURE TONES
 - FOR A PURE TONE TRAVELING ALONG "X", WE CAN INCLUDE THIS AS A PREFACTOR ON OUR WAVE:

$$\Delta p(x,t) \propto 10^{-\alpha x/20} \cos(2\pi \frac{f}{c}(x-ct))$$

RECALL THIS PREFACTOR IS THE CONVERSION FROM dB TO AMPLITUDE SCALE, WITH

αx = DIFFERENCE IN dB

DEMO PLOT THIS IN DESMOS FOR AIR: 5.3 PURE TONE DECAY

$c = 340$ m/s

f = SLIDER (220 Hz INITIALLY)

α = SLIDER (1 dB/M INITIALLY)

- WALK THROUGH / ANIMATE FIRST 3 CURVES IN ORDER

- GUESS + ANIMATE

- PLUG IN ACTUAL LOSS RATE FOR 220 Hz AT 300K, 50% HUMIDITY

$\alpha = 0.000745$ dB/m

- ZOOM OUT NOTE X-AXIS IN METERS

SANITY?

Q SO WHY ARE FAR AWAY NOISES QUIET?

- SO WHEN IS LOSS RELEVANT??

- NOTICE: WHEN CLOSE TO A SOURCE, 1M MEANS A LOT

WHEN FAR AWAY, ZOOM X TO ~ 1000 m ZOOM IN ON Y, PAN TO 1000m + ZOOM IN ON X

- DOES ± 1 m MAKE A DIFFERENCE HERE?

NOT LIKE WHEN YOU'RE CLOSE!

ABSORPTION DEPENDS ON FREQUENCY!

- GENERALLY, HIGH FREQUENCIES ARE ABSORBED FASTER

- ENABLE CURVE 6

- THIS IS 2200 Hz WITH THE SAME STARTING AMPLITUDE. $\lambda \approx 0.042966$

$\Rightarrow$ AT 1 km, 220 Hz WILL DOMINATE.

- ROOM BACK HOME, NOTING 220 Hz PULLS AWAY FROM THE $\frac{1}{R^2}$ ENVELOPE SOONER.

- ENABLE CURVE 7 (SUM, WITHOUT $\frac{1}{R^2}$)

10pt

Q AS WE TRAVEL ALONG X, WHAT WILL THIS WAVEFORM DO?

BONUS: SIMULATE THE DISTANCE-DEPENDENCE OF A THUNDER CLAP.

1. FIND A "BANG" SOUND

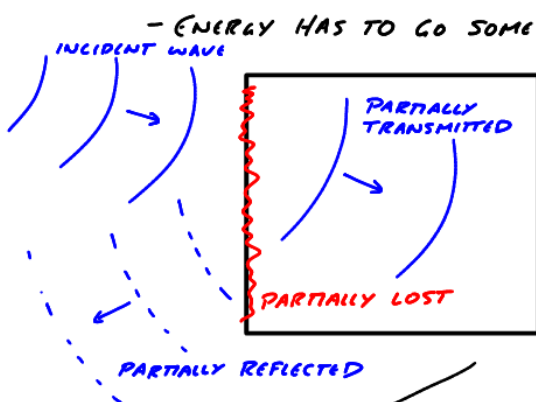
2. DECOMPOSE INTO PURE TONES

3. APPLY SOUND ABSORPTION AT ONE DISTANCE

4. RECONSTRUCT

(NOTE I AM HAPPY TO HELP W/ ALL BONUS QUESTIONS)

LOSS AT A SURFACE



- ENERGY HAS TO GO SOMEWHERE! OPTIONS:

1. REFLECTION (ECHO)

$\sim$ BIGGEST FOR DENSE, HARD, FLAT SURFACES

2. TRANSMISSION INTO MATERIAL WAVES

$\sim$ BIGGEST FOR "MATCHED" MATERIALS

3. "LOSS" AT SURFACE

$\sim$ BIGGEST FOR LIGHT, SOFT, POROUS SURFACES

- GENERALLY MORE ABSORPTION AT HIGHER FREQUENCIES

- ALL DEPEND ON FREQUENCY IN GENERAL

- CAN GET COMPLICATED IN A HURRY!

FOAM TO ABSORB SOUND
ROCK TO REFLECT

(RECORDING STUDIO, MY LAB)
(CANYON ECHOS)

- FINALLY, CONSERVATION OF ENERGY LEADS TO

INTENSITY
SUMMATION RULE
AT A SPOT ON A
SURFACE

$$I_{\text{TRANSMITTED}} + I_{\text{REFLECTED}} + I_{\text{LOST}} = I_{\text{INCIDENT}}$$

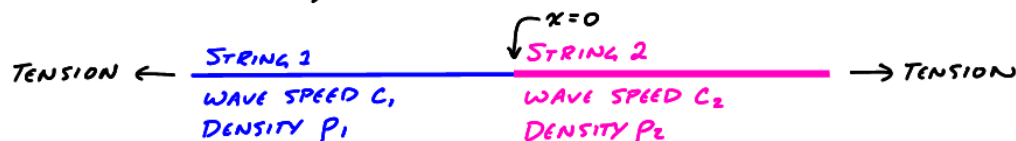
SO WHAT CAN WE SAY ABOUT
THESE TWO?

REFLECTION & TRANSMISSION

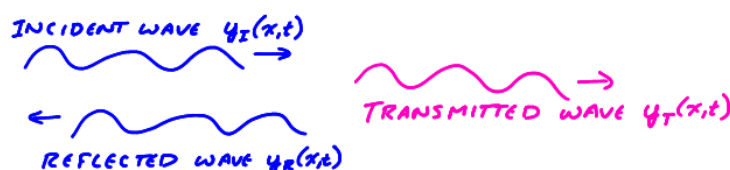
HOW WAVES INTERACT WITH BOUNDARIES

IDEAL STRING

- TO BUILD INTUITION, LET'S CONSIDER THE SIMPLEST WAVE SYSTEM: IDEAL STRINGS



- IF WE SEND A WAVE FROM THE LEFT, WHAT WILL HAPPEN? (IGNORE LOSS!)



- AGAIN IT IS EASIEST TO CONSIDER A PURE TONE FIRST! SO HOW MUCH DO WE KNOW?

SANITY

Q CAN THE REFLECTED AND TRANSMITTED WAVE HAVE FREQUENCIES DIFFERENT FROM THAT OF THE INCIDENT WAVE?

END OF
WED OCT 9

LEAVE
UP.

CONSTRAINS SOLUTION!

$$\Rightarrow y_i(x,t) = H_i \cos[2\pi(\underbrace{\lambda_1 x - ft}_{=f/c_1})] \leftarrow \text{ASSUMED INPUT (ALL KNOWN)}$$

$$y_r(x,t) = H_r \cos[2\pi(-\lambda_1 x - ft) + \phi_r] \leftarrow \text{SWITCHES DIRECTION}$$

$$y_t(x,t) = H_t \cos[2\pi(\underbrace{\lambda_2 x - ft}_{=f/c_2}) + \phi_t]$$

NOTE:

- WE "KNOW" H_i, f, c_1

- WE WISH TO SOLVE FOR H_r, ϕ_r, H_t, ϕ_t

- FURTHER CONSTRAINTS:

1. STRING DOES NOT BREAK $\Rightarrow y_i(0,t) + y_r(0,t) = y_t(0,t)$

STRING 1
STRING 2

$$\Rightarrow H_i \cos(2\pi ft) + H_r \cos(2\pi ft + \phi_r) = H_t \cos(2\pi ft + \phi_t)$$

- ONE WAY TO MAKE THIS VALID FOR ALL t IS TO ALIGN PHASES:

e.g.: $\phi_r = \phi_t = 0$ (ONE PATH TO THE SOLUTION)

$$\Rightarrow H_i + H_r = H_t \quad (1)$$

KNOWN $\uparrow$ UNKNOWN $\uparrow$

2. STRINGS HAVE NO KINKS $\Rightarrow$ SLOPE OF STRING 1 = SLOPE OF STRING 2 AT BOUNDARY

- A TRUE KINK WILL UNKINK INSTANTLY (FUN DISCUSSION)

ENTHUSIASTS: NO KINK $\Rightarrow \partial_x y_i(0,t) + \partial_x y_r(0,t) = \partial_x y_t(0,t)$

$$\Rightarrow -H_i \frac{2\pi f}{c_1} \sin(2\pi ft) + H_r \frac{2\pi f}{c_1} \sin(2\pi ft) = -H_t \frac{2\pi f}{c_2} \sin(2\pi ft)$$

$$\Rightarrow \frac{H_i}{c_1} - \frac{H_r}{c_1} = \frac{H_t}{c_2} \quad (2) \quad \text{WAVE SPEED STRETCHES THE WAVE HORIZONTALLY} \Rightarrow \text{SLOPE} \propto \frac{1}{c}$$

$\Rightarrow$ 2 EQUATIONS & 2 UNKNOWN! :)

$$\textcircled{1} \rightarrow \textcircled{2} \Rightarrow -\frac{H_I}{C_1} + \frac{H_R}{C_1} = -\frac{H_I + H_R}{C_2}$$

$$\Rightarrow -H_I C_2 + H_R C_2 = -H_I C_1 - H_R C_1$$

$$\Rightarrow H_R = \frac{C_2 - C_1}{C_2 + C_1} H_I$$

$$\textcircled{1} \Rightarrow H_I + \frac{C_2 - C_1}{C_2 + C_1} H_I = H_T$$

$$\Rightarrow H_T = \frac{2C_2}{C_2 + C_1} H_I$$

LEAVE
UP

REMEMBER THE "SHAPE"
OF THESE EQUATIONS FOR
LATER :)

(THIS IS CLOSE TO, BUT
NOT QUITE, A BLUE BOX
EQUATION.)

30pt

Q

IF THE INCIDENT WAVE CARRIES (AVERAGE) POWER P_I , HOW MUCH
POWER IS REFLECTED? TRANSMITTED?

10pt

Q

WHY IS THIS NOT JUST $\left(\frac{H_T}{H_I}\right)^2$?

DEMO 5.4 WAVE AT A BOUNDARY (DESMOS)

- SHOW INCIDENT

- SHOW REFLECTION, NOTING WAVELENGTH + AMPLITUDE

10pt

Q

CAN REFLECTED AMPLITUDE EXCEED INCIDENT AMPLITUDE?

- SLIDE C_2 AROUND + SHOW THIS

20pt

Q CAN TRANSMITTED AMPLITUDE EXCEED INCIDENT AMPLITUDE?

10pt

Q USE INTUITION: WHAT KIND OF STRING 2 MIGHT ACHIEVE THIS?

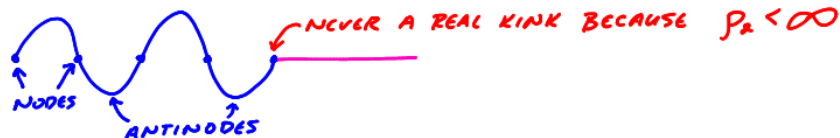
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Q SUPPOSE THE SECOND STRING IS ABSURDLY HEAVY THIS MEANS:

$$C_1 \gg C_2 \quad H_R \rightarrow$$

$$C_1 \ll C_2 \quad H_T \rightarrow$$

$$C_1 = C_2$$



$\Rightarrow$ "CLAMPED" BOUNDARY CONDITION
(OR "FIXED" OR "NODE")

- SHOW IN DESMOS: $C_2 \rightarrow 0.0001$

- ZOOM IN ON KINK NOTING THE WAVELENGTH + CONTINUITY

- THIS COULD BE A TOY MODEL FOR SOUND RADIATION AT THE CLAMPING POINT OF A PIANO STRING, E.G.

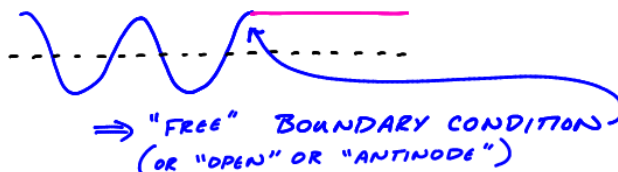
Q WHAT IF THERE IS ANOTHER CLAMP?

10pt

Q WHAT IF $\rho_2 \rightarrow 0$?

$$C_1 \quad C_2 \quad H_R \rightarrow$$

$$H_T \rightarrow$$

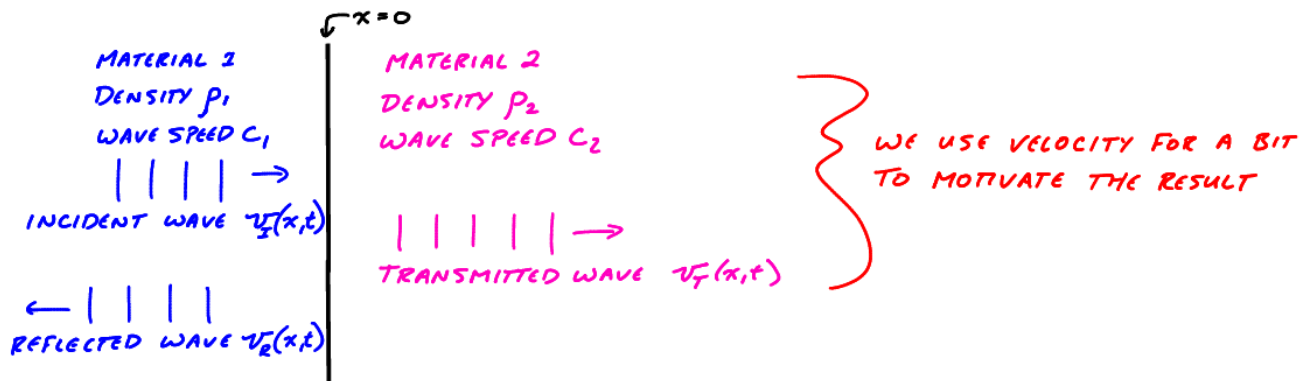


- SHOW WITH $C_2 = 10,000$ OR SO

- EASY TO REALIZE WITH AIR IN A WINDSTRUMENT!

SOUND WAVES

- WHAT HAPPENS TO SOUND WAVES?



- THIS TIME WE ONLY MOTIVATE THE FORM OF THE TRANSMISSION & REFLECTION

1. IMAGINE A LITTLE CUBE OF MATERIAL 1 FLYING AT A CUBE OF MATERIAL 2 (SAME SIZE)

BEFORE:



AFTER:



"TILDE" IS JUST LIKE A SUBSCRIPT, MEANING "AFTER" IN THIS CASE. SO $\tilde{v}_1$ IS A DIFFERENT VALUE THAN v_i .

2. WHAT HAPPENS WHEN THEY COLLIDE?

10pt

Q WHAT IS THE KINETIC ENERGY OF EACH CUBE BEFORE COLLIDING?

10pt

Q WHAT IS THE TOTAL MOMENTUM BEFORE COLLIDING?

- FUN FACTS:

- TOTAL MOMENTUM IS ALWAYS CONSERVED

$$\textcircled{1} \Rightarrow \underbrace{m_1 v_i}_{\text{BEFORE COLLISION}} = \underbrace{m_1 \tilde{v}_1 + m_2 \tilde{v}_2}_{\text{AFTER COLLISION}}$$

- IF WE IGNORE DISSIPATION/LOSS TOTAL KINETIC ENERGY IS ALSO CONSERVED

$$\textcircled{2} \Rightarrow \underbrace{\frac{1}{2} m_1 v_i^2}_{\text{BEFORE}} = \underbrace{\frac{1}{2} m_1 \tilde{v}_1^2 + \frac{1}{2} m_2 \tilde{v}_2^2}_{\text{AFTER}}$$

10pt

Q WHAT ARE THE UNKNOWN IN THESE EQUATIONS?

- TWO SOLUTIONS:

1. $\tilde{v}_1 = v_1$ AND $v_2 = 0$ (NO COLLISION)

2. $\tilde{v}_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1 = \frac{p_1 - p_2}{p_1 + p_2} v_1$ AND $\tilde{v}_2 = \frac{2m_1}{m_2 + m_1} v_1 = \frac{2p_1}{p_2 + p_1} v_1$

"SHAPE" OF EQUATION FAMILIAR?

- THIS IS FOR PARTICLES COLLIDING, JUST TO MOTIVATE THE RELEVANCE OF p FOR COMPRESSION WAVES.

BONUS: BE THIS! DO THIS!

- IF "ONE" IS "CAREFUL", FOR SOUND WAVES, IT TURNS OUT

REFLECTED INTENSITY

$$I_R = \left(\frac{p_1 c_1 - p_2 c_2}{p_1 c_1 + p_2 c_2} \right)^2 I_I$$

INCIDENT INTENSITY

TRANSMITTED INTENSITY

$$I_T = \frac{4(p_1 c_1)(p_2 c_2)}{(p_1 c_1 + p_2 c_2)^2} I_I$$

SAME "SHAPE" AS FOR THE STRING, BUT $c \rightarrow pc \equiv "Z"$

- PEOPLE DEFINE THE "IMPEDANCE" Z FOR SOUND WAVES:

$$Z = \rho c \quad (\text{A THING THAT ALWAYS APPEARS AS A GROUP})$$

EVERY MATERIAL HAS ONE. $\ddot{O}$
(MORE BELOW)

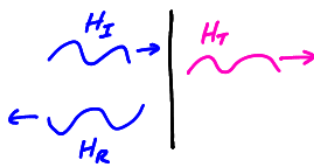
END OF
OCT 21

GENERALIZATION TO OTHER WAVE MEDIA

- THE "SHAPE" OF THESE EQUATIONS IS THE SAME FOR ALL WAVES I HAVE SEEN

- SOUND
- LIGHT
- COAXIAL CABLES
- WATER
- QUANTUM PARTICLES

IN GENERAL, AT A BOUNDARY,



$$H_R = \frac{Z_2 - Z_1}{Z_2 + Z_1} H_I$$

$$H_T = \frac{2Z_2}{Z_2 + Z_1} H_I$$

FOR STRINGS, THESE ARE DISPLACEMENT AMPLITUDES
FOR SOUND, THESE ARE PRESSURE AMPLITUDES

REFLECTED AMPLITUDE
"IMPEDANCE" OF MEDIUM 1
"IMPEDANCE" OF MEDIUM 2

TRANSMITTED AMPLITUDE
FOR "IMPEDANCES" $Z_1 + Z_2$

(JUST TO GET THE (-) SIGN RIGHT ON THE REFLECTED WAVE. FOR PRESSURE, $B_i \rightarrow -B_i$)

- POWER OR INTENSITY EQUATIONS ARE NOT JUST THE SQUARE:

$$I_R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2 I_I$$

REFLECTED INTENSITY

$$I_T = \frac{4Z_2 Z_1}{(Z_2 + Z_1)^2} I_I$$

TRANSMITTED INTENSITY (NEW MATERIAL)

- FOR IDEAL STRING, THE IMPEDANCE IS JUST THE SPEED

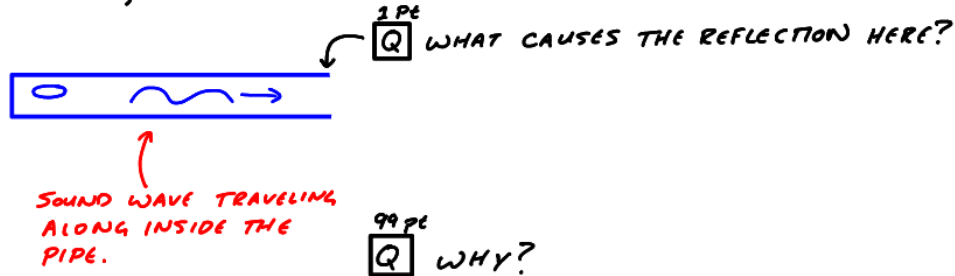
$Z = c$ IDEAL STRING WAVE IMPEDANCE (OR PROPORTIONAL TO IT $\ddot{u}$)

- FOR SOUND IN AIR (WITHOUT PROOF):

$Z = \rho c$ MATERIAL IMPEDANCE $\left[\frac{\text{kg}}{\text{m}^2 \text{s}} \right]$ FOR "FREE" WAVES (NOT CONFINED BY PIPES, ETC)

(Annotations: ρ is MASS DENSITY, c is SPEED OF SOUND)

- IN A FLUTE, SOUND WAVES REFLECT FROM THE OPEN END



20pt

Q MATCHING GAME:

MAXIMUM REFLECTION

MAXIMUM TRANSMISSION

$Z_2 \gg Z_1$

$Z_2 > Z_1$

$Z_2 = Z_1$

$Z_2 < Z_1$

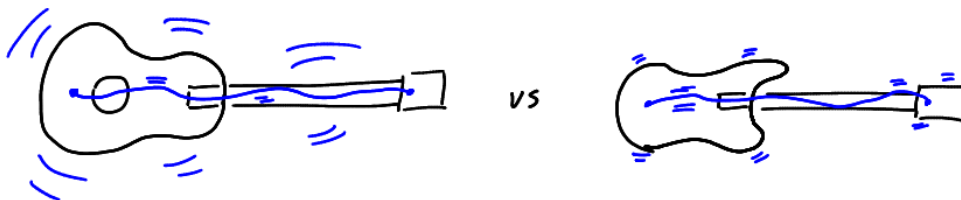
$Z_2 \ll Z_1$

- APPLICATION TO INSTRUMENTS:

⇒ METAL HORNS CONTAIN SOUND INSIDE PIPES BY IMPEDANCE MISMATCH



⇒ ACOUSTIC GUITARS ARE LOUDER THAN ELECTRIC BECAUSE THE STRINGS ARE BETTER IMPEDANCE MATCHED TO THE BODY (AND THE BODY IS BETTER MATCHED TO THE AIR) ^{WITH NO AMP $\ddot{u}$}



1pt

Q WHICH STRING WILL RING LONGER?

19pt

Q WHY?

BONUS: MEASURE THIS EFFECT! (ELECTRIC GUITAR + TABLE?)

- SENSE OF SCALE

MATERIAL	DENSITY	WAVE SPEED	IMPEDANCE
AIR	1.3 kg/m^3	340 m/s	$440 \frac{\text{kg}}{\text{m}^2\text{s}}$
WATER	1000 kg/m^3	1500 m/s	$1.5 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
BONE	1908 kg/m^3	4080 m/s	$7.8 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
PLASTIC			$\sim 3 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
GLASS			$\sim 15 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
ALUMINUM			$\sim 17 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
COPPER			$\sim 42 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
ALUMINA			$\sim (21 \text{ to } 45) \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$
TUNGSTEN			$104 \times 10^6 \frac{\text{kg}}{\text{m}^2\text{s}}$

20pt

Q

IF ^{AIR} SOUND LANDS ON WATER WITH AN INTENSITY OF 1 W/m^2
HOW MUCH IS TRANSMITTED? (SET UP PROBLEM + MAYBE ESTIMATE)

10pt

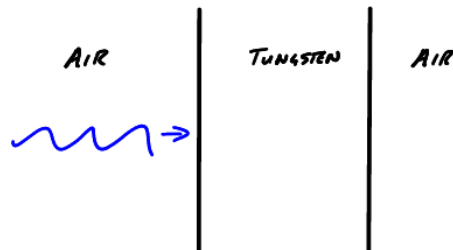
Q

HOW MUCH REFLECTS?

20pt

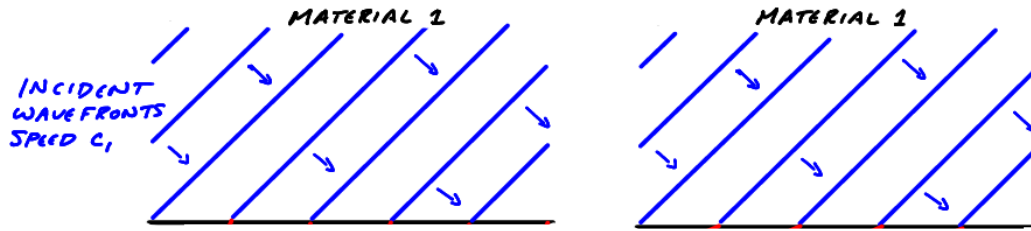
Q

HOW MUCH WOULD TRANSMIT THROUGH 10m OF TUNGSTEN?
(IGNORE LOSS + INTERFERENCE)



REFRACTION

SPEAKING OF WAVES PASSING FROM ONE MEDIUM TO ANOTHER
CONSIDER THIS:



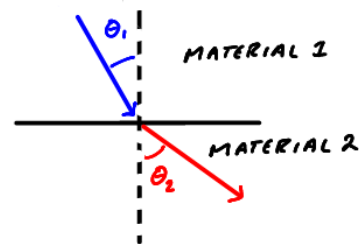
soft



WHAT DO THE TRANSMITTED WAVEFRONTS LOOK LIKE?

GEOMETRY $\Rightarrow$ SNELL'S LAW

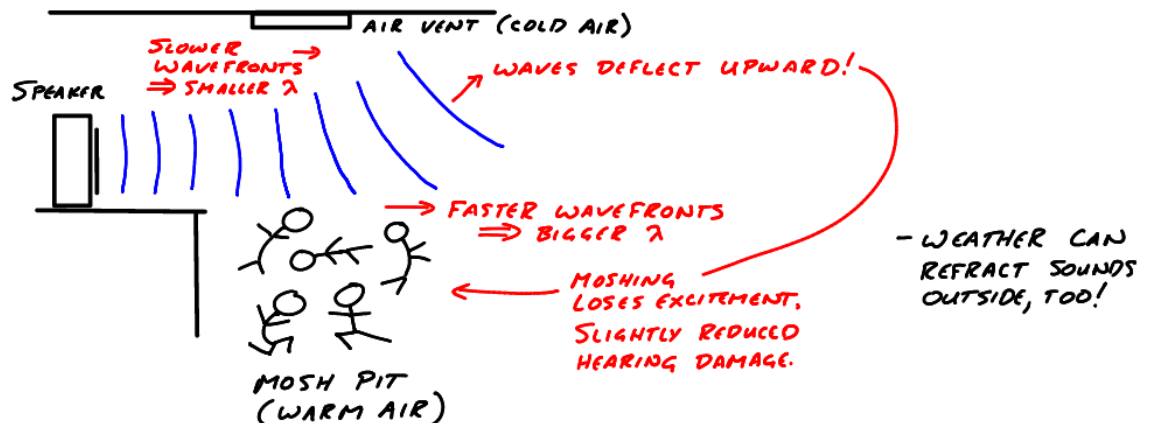
$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{c_1}{c_2}$$



- TRUE FOR ALL WAVE PHENOMENA

- SOUND
- WATER
- LIGHT
- QUANTUM PARTICLES

- MATERIALS CAN BE AIR AT DIFFERENT TEMPERATURES!



BONUS: ESTIMATE THIS EFFECT :)

- NOTE THE REFLECTED WAVE HAS THE SAME SPEED $\Rightarrow$ NO ANGLE CORRECTION
 $\Rightarrow$ ANGLE IN = ANGLE OUT

SUMMARY

WAVELENGTH ETC...

$$f \equiv \frac{1}{T}$$

IN TIME (FREQUENCY, UNITS $\frac{1}{s} = \text{Hz}$)

$$k \equiv \frac{1}{\lambda}$$

IN SPACE ("SPATIAL FREQUENCY", UNITS $\frac{1}{m}$)

$$c = \frac{\lambda}{T} = \lambda f = \frac{f}{k} = \frac{1}{Tk}$$

WAVE SPEED (JUST USE UNITS!)

Wavelength [m] → λ
 Frequency [Hz] → f
 Period [s] → T
 Spatial Frequency [$\frac{1}{m}$] → k

DIFFRACTION

$$\theta_{\text{slit}} \approx \frac{\lambda}{W}$$

SPREAD ANGLE FOR PLANE WAVES LEAVING A CONFINED SOURCE (DIFFRACTION ANGLE)

Wavelength → λ
 Width of source / confinement → W
 $\pm \theta_{\text{slit}}$ CONTAINS MOST OF THE POWER

LOSS

$$I(\Delta R) \propto 10^{-\alpha \Delta R / 10}$$

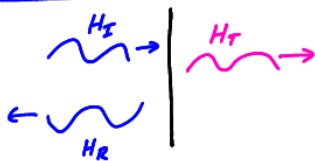
LOSS DECAY FACTOR

"LOSS RATE" IN dB/m → α
 Distance traveled → ΔR

INTENSITY SUMMATION RULE AT A SPOT ON A SURFACE

$$I_{\text{transmitted}} + I_{\text{reflected}} + I_{\text{lost}} = I_{\text{incident}}$$

REFLECTION + TRANSMISSION



$$H_R = \frac{z_2 - z_1}{z_2 + z_1} H_I$$

REFLECTED AMPLITUDE

"IMPEDANCE" OF MEDIUM 1 → z_1
 "IMPEDANCE" OF MEDIUM 2 → z_2

$$H_T = \frac{2z_2}{z_2 + z_1} H_I$$

TRANSMITTED AMPLITUDE

FOR "IMPEDANCES" z_1, z_2

FOR STRINGS, THESE ARE DISPLACEMENT AMPLITUDES
 FOR SOUND, THESE ARE PRESSURE AMPLITUDES!
 (MORE LATER!)

- POWER OR INTENSITY EQUATIONS ARE NOT JUST THE SQUARE!

$$I_R = \left(\frac{z_2 - z_1}{z_2 + z_1} \right)^2 I_I$$

REFLECTED INTENSITY

$$I_T = \frac{4z_1 z_2}{(z_2 + z_1)^2} I_I$$

TRANSMITTED INTENSITY (NEW MATERIAL)

$$z = c$$

IDEAL STRING WAVE IMPEDANCE (OR PROPORTIONAL TO IT $\ddot{u}$)

$$z = \rho c$$

MATERIAL IMPEDANCE [$\frac{\text{kg}}{\text{m}^2 \text{s}}$]
 (NOT QUITE A BLUE BOX YET.. MORE LATER!!)

Speed of sound → c
 Mass density → ρ

REFRACTION

SNELL'S LAW

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{c_1}{c_2}$$

